

S.No. : 316

NCS 4401

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43209

**Roll
No.**

B. Tech. Examination, 2026

(Even Semester)

DATABASE MANAGEMENT SYSTEMS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) Define database system.
- (b) What are the advantages of DBMS?
- (c) What is a weak entity?
- (d) Explain derived attribute.

[P. T. O.

- (e) ✓ What is aggregate functions?
- (f) ✓ What is the purpose of the WHERE clause?
- (g) ✓ Define ACID property.
- (h) Define checkpoints in recovery.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$
- (a) Explain different types of database users and their roles in DBMS.
 - (b) ✓ Describe specialization and generalization in the enhanced ER model with examples.
 - (c) ✓ Explain Boyce-code normal form and how it differs from third normal form.
 - (d) Explain recoverable and non-recoverable schedules with examples.

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) ✓ Explain three tier DBMS architecture with neat diagram.
 - (b) ✓ What is language in DBMS? Explain DDL and DML language.
 - (c) Explain data abstraction and discuss its levels with suitable examples.
4. (a) ✓ What are the mapping constraints in an ER-model? Explain different types of mapping constraints.
- (b) Explain operations of relational algebra :
 - (i) Selection
 - (ii) Projection
 - (iii) Joins
 - (c) ✓ Define concepts of super key, candidate key, primary key and foreign key.
5. (a) Consider a student table (Roll no., Name, Marks, Dept.) :
- (i) Write a query to create a student table.
 - (ii) Display the student's detail, who's scoring highest marks in a table.

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- (iii) Write a subquery to find students whose marks are above average.
- (b) Define anomaly and different types of anomalies.
- ✓ (c) Given a relation R (A, B, C, D, E, F) with functional dependencies :

$$A \rightarrow BC$$

$$CD \rightarrow E$$

$$B \rightarrow D$$

$$E \rightarrow F$$

Find all candidate keys of relation R and also list all prime and non-prime attributes of relation R.

6. (a) ✓ What is transaction? Describe different transaction states with the help of a state transition diagram.
- ✓ (b) Explain strict two-phase locking (strict-2PL) and its advantages.
- (c) Describe deadlock detection and recovery techniques.
